



PLANT DISEASE DETECTION USING CNN AND TRANSFER LEARNING

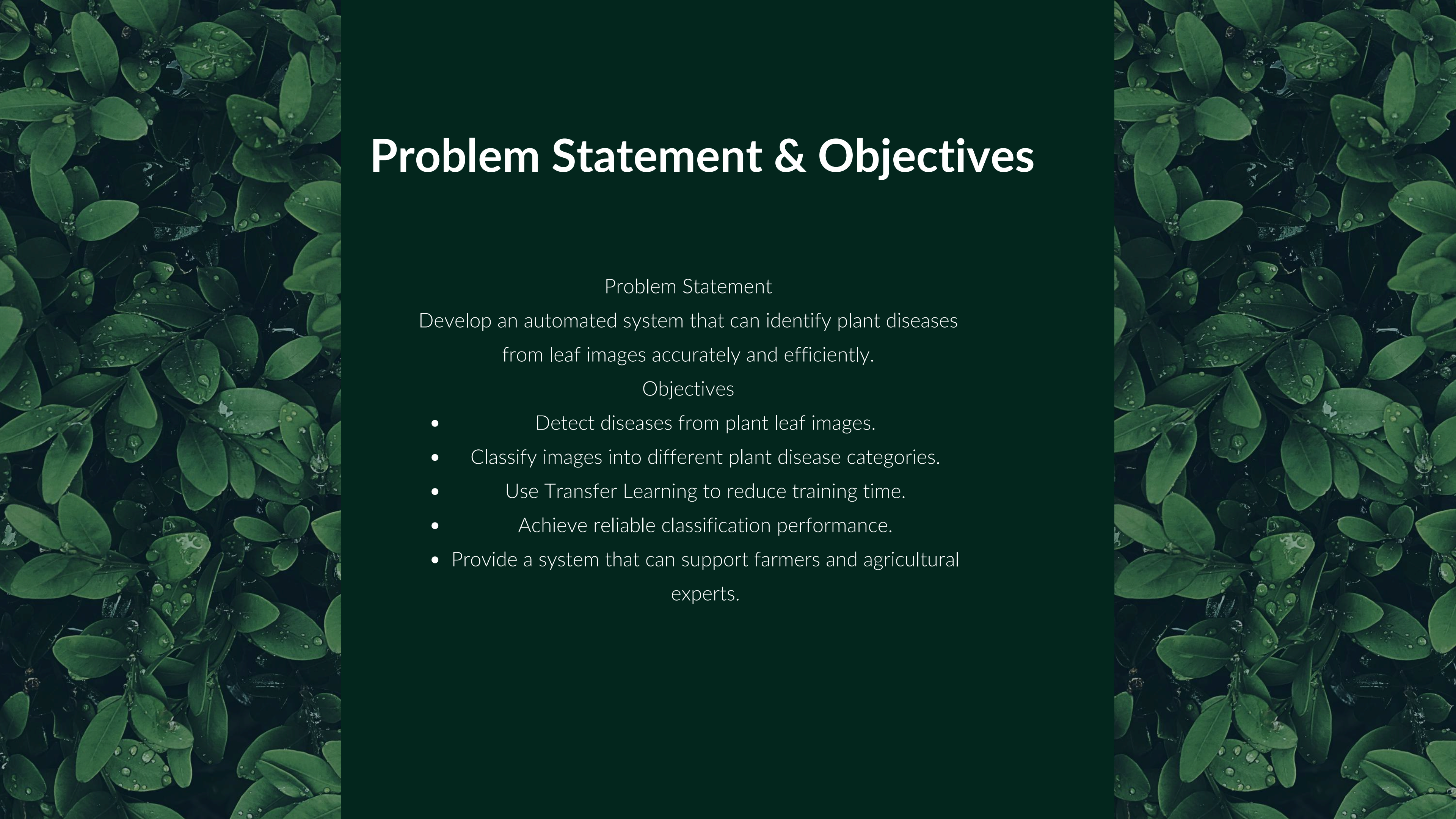
Ratna priya - 2520030156
Chandra Hasini - 2520030086



Introduction |

- Plant diseases can reduce crop quality and agricultural productivity.
- Traditional disease identification mainly depends on manual observation and expert knowledge.
- Manual identification can be time-consuming and difficult when diseases have similar symptoms.
- Deep Learning can automatically identify disease patterns from plant leaf images.
- Our project uses CNN and Transfer Learning for automated plant disease classification.
- Main Idea
- Leaf Image → Deep Learning Model → Disease Classification





Problem Statement & Objectives

Problem Statement

Develop an automated system that can identify plant diseases from leaf images accurately and efficiently.

Objectives

- Detect diseases from plant leaf images.
- Classify images into different plant disease categories.
- Use Transfer Learning to reduce training time.
- Achieve reliable classification performance.
- Provide a system that can support farmers and agricultural experts.

Dataset



PlantVillage Dataset

- Publicly available plant disease image dataset.
- Contains images from multiple crops and disease categories.
- Our project uses the 38-class plant disease classification problem.
- Images are used to train the model to recognize disease-specific visual patterns.
- Important features include:
 - Leaf color
 - Texture
 - Shape
 - Spots
 - Lesions

The literature review associated with the project also identifies PlantVillage as a commonly used benchmark dataset containing 38 crop-disease classes across 14 crops.

Plant_Disease_Literature_Review...



Proposed Methodology

Workflow

Dataset Collection



Image Preprocessing



Data Augmentation



MobileNetV2 Transfer Learning



Model Training



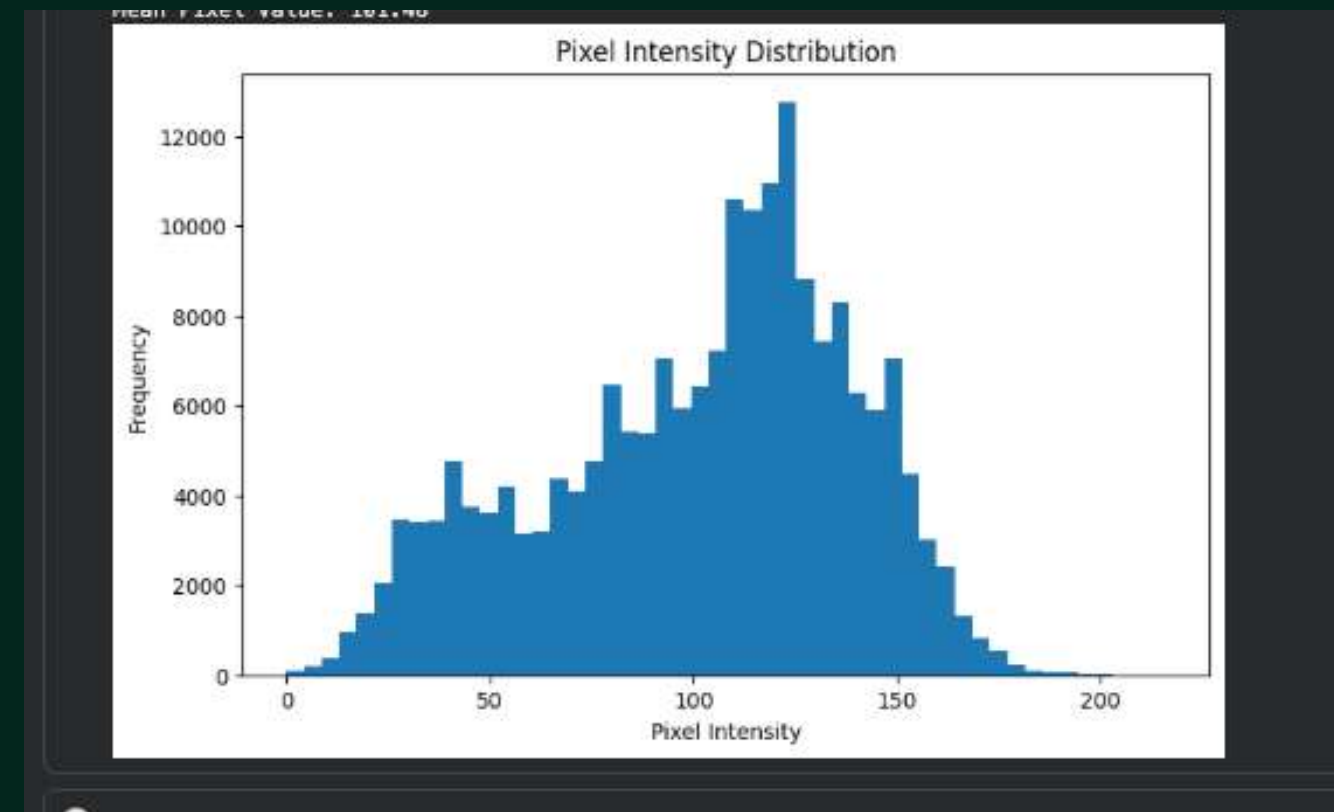
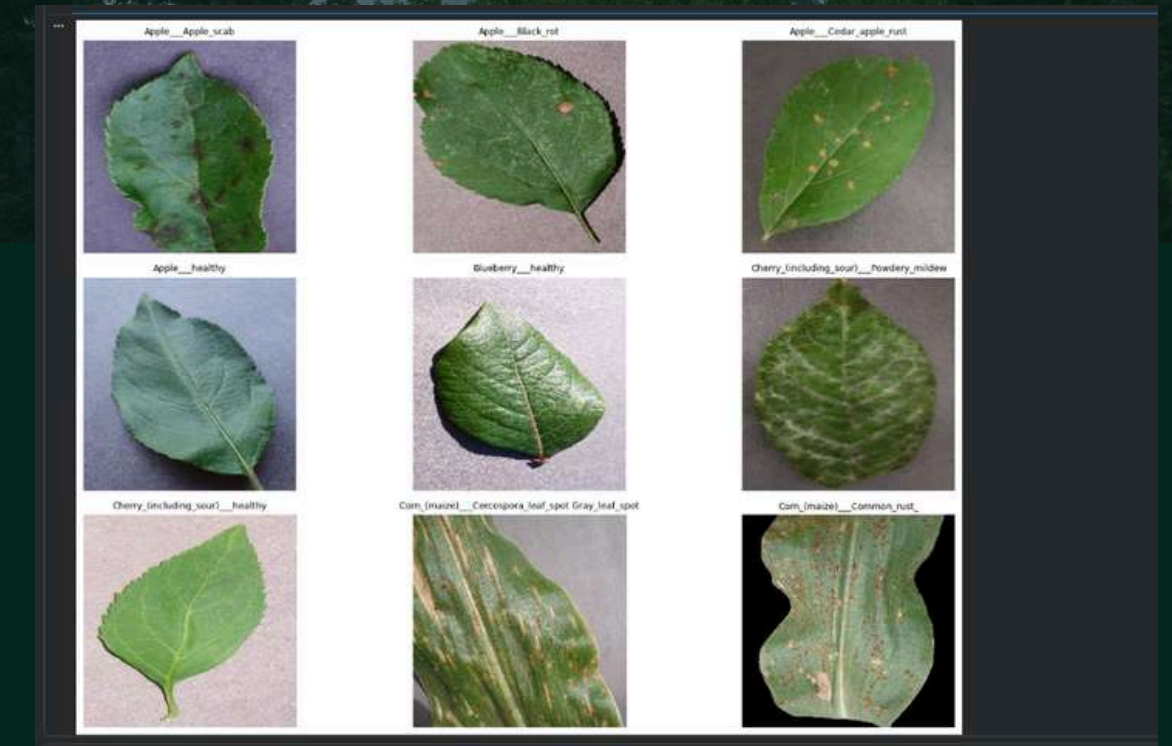
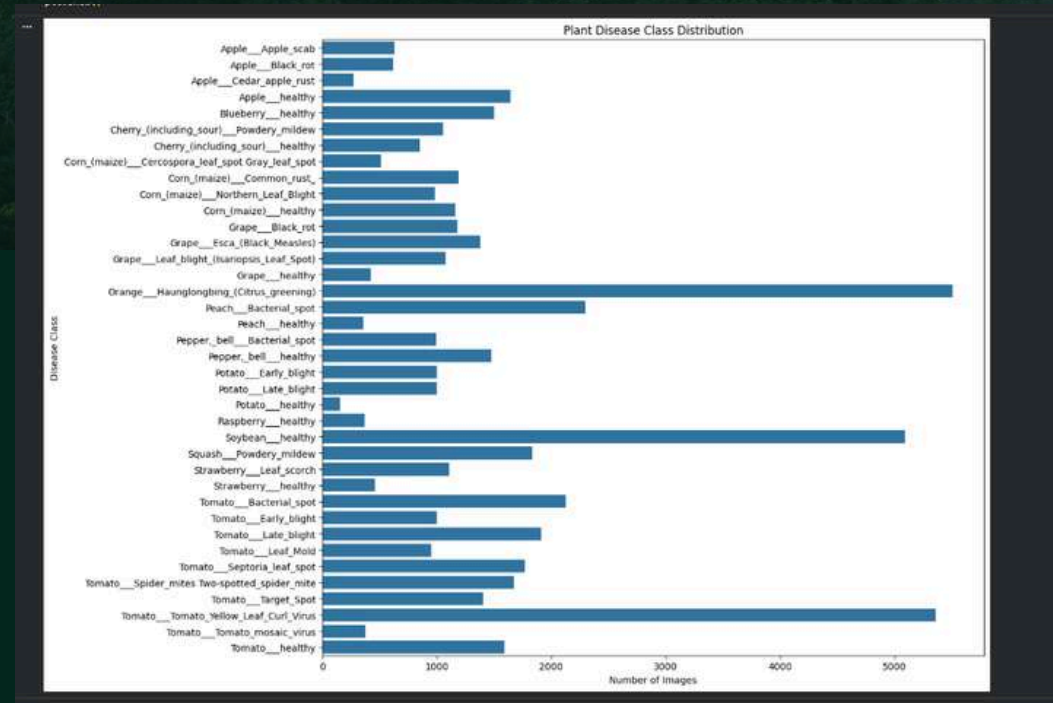
Model Evaluation



Disease Prediction

Preprocessing

- Resize images to the required input size.
- Normalize image pixel values.
- Apply augmentation to improve model generalization.



CNN & Transfer Learning

CNN

Convolutional Neural Network automatically learns important visual features from images.

It can identify:

- Edges
- Shapes
- Textures
- Spots
- Disease patterns

Transfer Learning

Instead of training a CNN completely from the beginning, we use a pre-trained MobileNetV2 model.

Advantages:

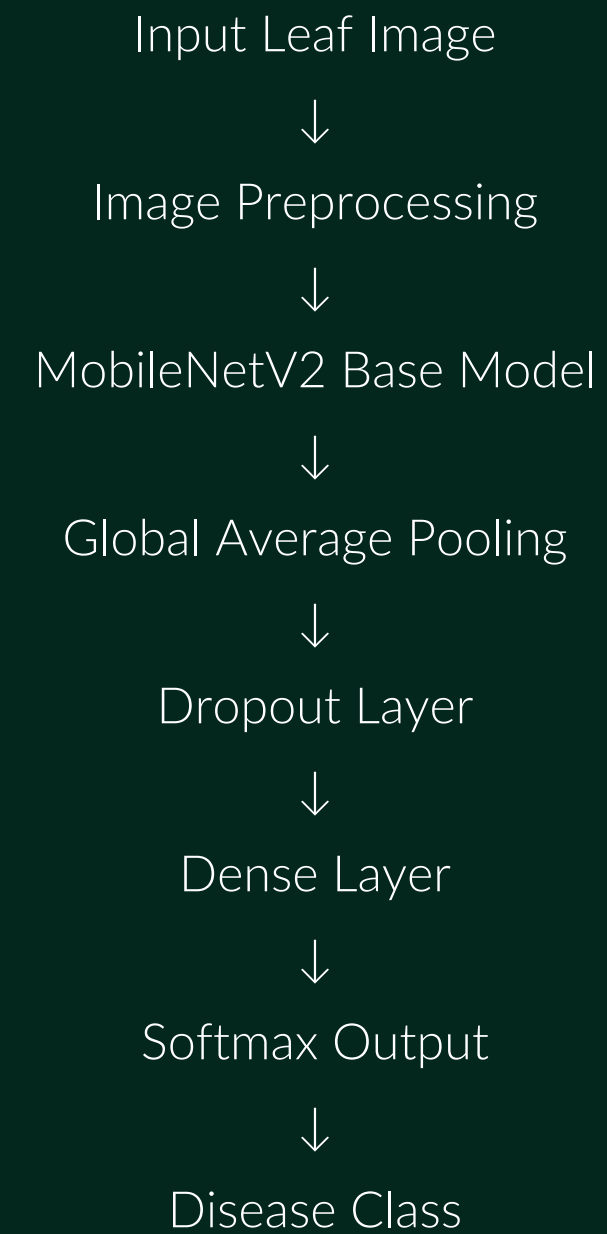
- Faster training
- Requires less computational resources
- Uses previously learned image features
- Suitable for image classification tasks

Our project uses MobileNetV2 with ImageNet-pretrained weights and a classification layer for the plant disease classes.



Model Architecture

Our Model



Training Configuration

- Model: MobileNetV2
- Optimizer: Adam
- Learning Rate: 0.001
- Loss Function: Sparse Categorical Crossentropy
- Output: 38 disease/healthy classes

Training & Evaluation

Training

The model was trained using the PlantVillage image dataset. During training, the model learned visual patterns associated with different plant diseases.

Evaluation Metrics

We evaluated the trained model using:

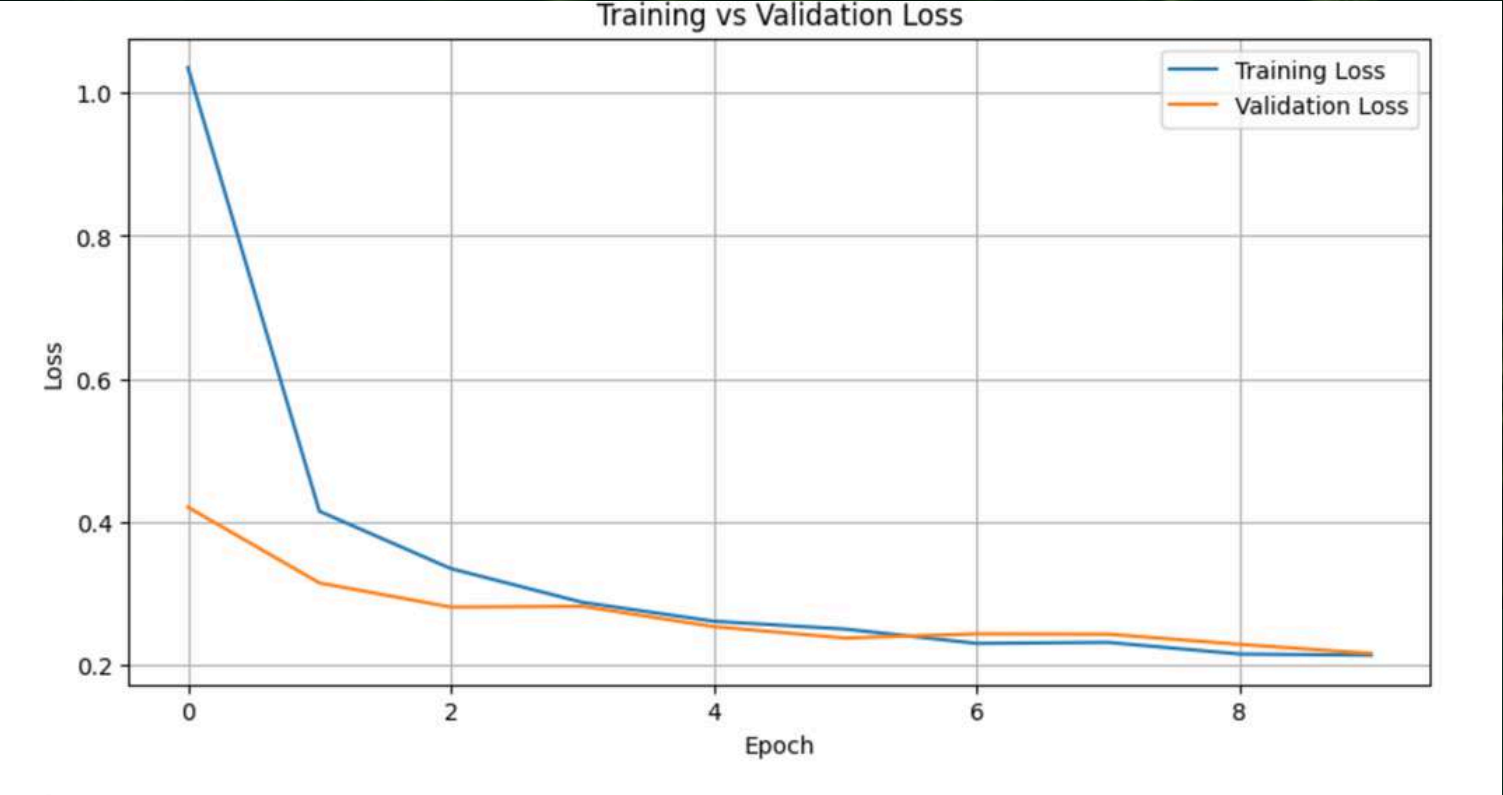
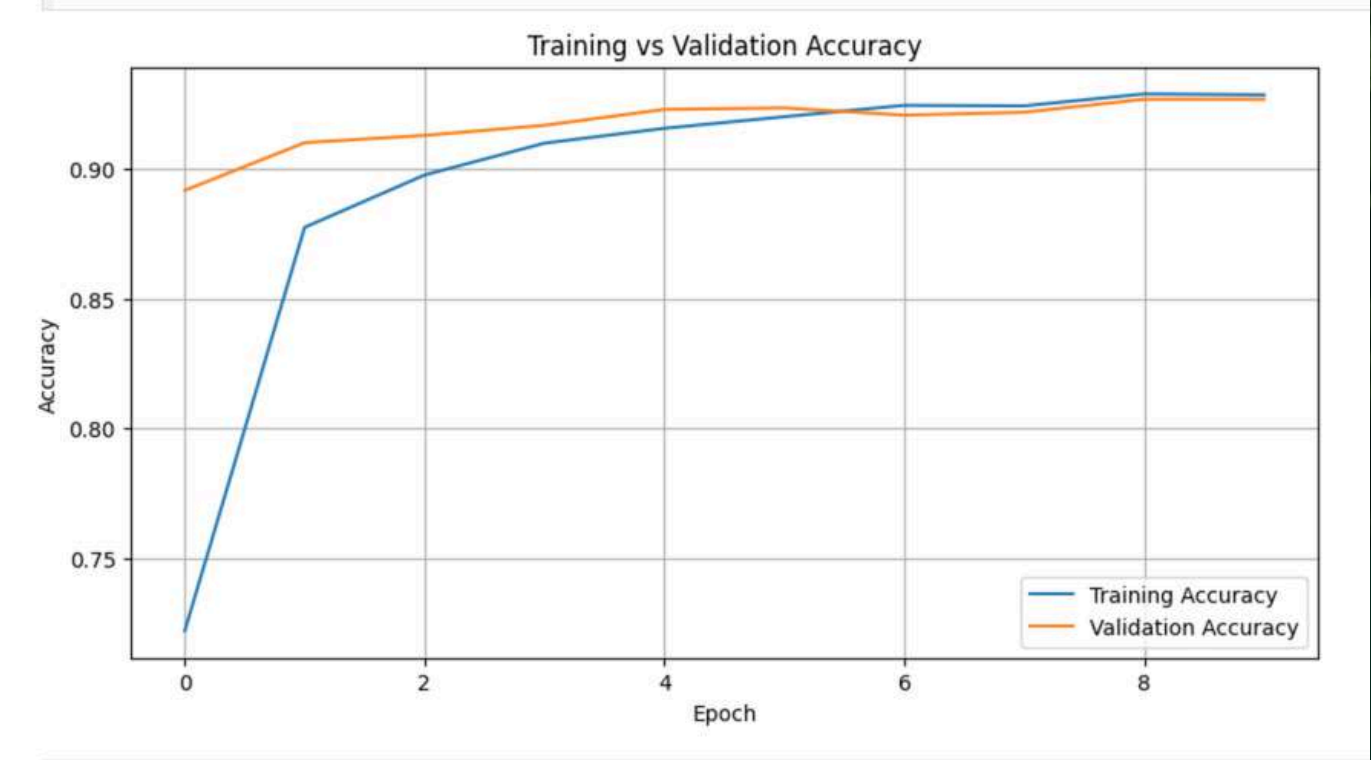
- Accuracy

- Loss
- Classification performance
- Confusion Matrix
- Class-wise prediction analysis

The project report specifies accuracy, precision, recall and F1-score as relevant evaluation measures for the system.



Results



Uploaded Plant Leaf



```
=====
PLANT DISEASE DETECTION RESULT
=====
Image: saved-image-file_00000000948c81f5b58a7552eea25240.jpg
Predicted Disease: Corn_(maize)___healthy
Confidence: 74.62%
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An aerial photograph of a dense, lush green forest. A paved road with white lane markings runs horizontally across the middle of the image, separating the forest into two sections. The trees are thick and vibrant green, filling the entire background.

Conclusion & Future Scope

Conclusion

- Developed a plant disease detection system using CNN and Transfer Learning.
- Used MobileNetV2 for efficient image classification.
- The model achieved 92.29% test accuracy on the evaluated test set.
- The system can automatically analyze plant leaf images and predict disease categories.
- This approach can support faster preliminary disease identification.

Future Scope

- Test the model using real-world field images.
- Improve performance on different lighting and background conditions.
- Develop a mobile/web application.
- Add more crops and disease categories.
- Provide real-time disease detection and guidance



Thank You

FOR YOUR ATTENTION